THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 02

Full Syllabus (Chapters 1–14) | Class 9 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(d)** are collinear	O is the midpoint of AB
2	**(a)** -11	$-2(5)+(-1)=-11$
3	**(a)** division by zero (undefined)	division by 0 is not defined
4	**(d)** Incorrect; the term $2ab$ is missing	$(a+b)^2=a^2+2ab+b^2$
5	**(b)** 15 cm	$d=\sqrt{17^2-8^2}=15$
6	**(c)** 132 cm	$2\pi r=2\times\frac{22}{7}\times21=132$
7	**(a)** theoretical probability	Law of Large Numbers
8	**(b)** $\frac{2}{11}$	2 favourable out of 11 letters
9	**(c)** Prakrit metre and poetry	7th century CE, Vṛttajātisamuchaya
10	**(a)** each implies the other	both the proposition and its converse are true
11	**(b)** If the school is closed, then it is Sunday.	swap the ‘if’ part and the ‘then’ part
12	**(d)** 68	$\frac{70\times2+80\times3+60\times5}{10}=68$
13	**(a)** Both A and R are true and R is the correct explanation of A	so the units digit is written and 1 carried
14	**(a)** 28 cm	its sides are AC/2, BD/2, AC/2, BD/2
15	**(a)** 70°	adjacent angles of a parallelogram add up to 180°
16	**(b)** 1	k/3 = 2/6 = 4/12 gives k = 1
17	**(a)** 8	solve the pair first
18	**(b)** Both A and R are true but R is not the correct explanation of A	both true; the cuboid formula does not explain the pyramid formula
19	**(c)** 144 cm^3	$\frac{1}{3}\times48\times9=144$
20	**(c)** 256 cm^2	$2(lw+wh+lh)=2(60+20+48)=256$

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	Distance of $(-5,-12)$ is $\sqrt{25+144}=13$; distance of $(13,0)$ is 13. Yes, both lie on the circle of radius 13 centred at the origin.	1+1
22	Amount $=200+50m$. $200+50m=750\Rightarrow m=11$ matches.	1+1
23	$-850+1200-450=-100$: he is in debt of ₹100.	1+1
24	$bh=96\Rightarrow h=\frac{96}{12}=8$ cm.	1+1
25	15% (1), the midpoint, because equal quantities have equal weights (1).	1+1
26	Al-Khwārizmī (c. 820 CE) wrote on Indian numerals; the Latin form of his name, Algoritmi, gave algorismus → algorism → algorithm (2).	2
27	False (1). The diagonals of every rhombus are perpendicular; a rhombus with equal diagonals (or a right angle) is a square (1).	1+1
28	TSA $=2\pi rh+2\pi r^2=2\pi r(h+r)$ (1). $2\pi rh$ is the curved surface; $2\pi r^2$ is the two circular ends (1).	1+1

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	Converse: If $a+x=a+y$, then $x=y$ (1). Proposition: adding a to equals gives equals; converse: subtract a from both sides (1). Used while solving equations by balancing (1).	1+1+1
30	Fourth vertex $(2,5)$. Diagonal $=\sqrt{36+16}=2\sqrt{13}$ units.	1+2
31	Points: $(0,0),(4,2)$; $(0,0),(2,2)$; $(0,0),(2,4)$. All pass through the origin; larger a ⇒ steeper line; $a=1$ makes equal angles with both axes.	2+1

32	Differences $2, 3, 4$ — not constant ⇒ not an AP (1+). Ratios $3, 2, \frac{5}{3}$ — not constant ⇒ not a GP (1).	1+1+1
33	Adding: $20x + 20y = 140$, so $x + y = 7$ (1). Subtracting: $2x - 2y = 6$, so $x - y = 3$ (1). Then $x = 5$ and $y = 2$; check $55 + 18 = 73$ (1).	1+1+1
34	Diagonal BD splits the parallelogram into two congruent triangles (1), so $\text{ar}(ABD) = 42$ cm^2 (2).	1+2
35	Half-chord $= \sqrt{r^2 - d^2}$ (Baudhāyana–Pythagoras) (1); chord $= 2\sqrt{r^2 - d^2}$ (1). For $r = 10, d = 6$: $2\sqrt{64} = 16$ (1).	1+1+1
	OR: The angle at the origin is $90°$, so the hypotenuse joining $(6, 0)$ and $(0, 8)$ is a diameter (1). Its midpoint $(3, 4)$ is the centre (1); radius $= \frac{1}{2}\sqrt{36 + 64} = 5$ (1).	1+1+1
36	(i) $[1, 5, 25]$ (1). (ii) gcd $= 25$ (1). (iii) Lists are in increasing order, so the common list is too, and the largest common divisor is last (1).	1+1+1
	OR: $33 \bmod 7 = 5$ (1). Common divisors of 33 and 7: only 1 (1). Common divisors of 7 and 5: only 1; so the two sets of common divisors are the same, as claimed (1).	1+1+1

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	Adjacent angles add up to $180°$, so half of $\angle A$ plus half of $\angle B$ is $90°$ (1). In the triangle formed by AB and the bisectors of $\angle A$ and $\angle B$, the third angle is $180° - 90° = 90°$ (2). The same holds at each of the other three corners, so the figure PQRS formed has four right angles: a rectangle (1). If $AB = BC$ (a rhombus), the bisectors are the diagonals and all meet in one point, so no quadrilateral is formed (1).	1+2+1+1
	OR: (i) $40 + 50 + 30 + 240 = 360°$ (1). (ii) $360°$ (1). (iii) Rotate a copy through $180°$ about the midpoint of each edge of every copy added; the copies fit exactly (2). (iv) Each angle of a regular pentagon is $108°$; 360 is not a multiple of 108, so copies cannot fill a full turn around a point (1).	1+1+2+1
38	(i) $n = 7$: $49 + 2 = 51 = 3 \times 17$ is not prime (1½). (ii) $n = 2$ is prime but $4 - 1 = 3$ is not a multiple of 4 (1½). (iii) $n = 18$ is divisible by 9, not by 27 (2).	1½+1½+2
	OR: (i) The argument starts from 'odd number of factors' and ends at 'perfect square', i.e. the direction (odd factors ⇒ square), the converse (2). (ii) Start from $n = f \times f$; the pair (f, f) exists; two such repeated pairs cannot exist, so all other factors pair up in twos and the count is odd ⇒ proposition proved (3).	2+3
39	(i) HHH, HHT, HTH, THH, HTT, THT, TTH, TTT (2). (ii) 3 outcomes ⇒ $\frac{3}{8}$ (1). (iii) $1 - \frac{1}{8} = \frac{7}{8}$ (1). (iv) HHH, TTT ⇒ $\frac{2}{8} = \frac{1}{4}$ (1).	2+1+1+1
	OR: (i) A personal judgement, e.g. 'it is sunny so I think rain is unlikely'; different people may differ (2). (ii) Count the days with rain in past years and compute the relative frequency (2). (iii) Rain depends on many sensitive atmospheric factors, so the outcome cannot be predicted with certainty (1).	2+2+1
40	(i) $\frac{126}{10} = \frac{63}{5}$. (ii) $\frac{120}{10000} = \frac{3}{250}$. (iii) $\frac{3052}{1000} = \frac{763}{250}$. (iv) $\frac{23}{99}$. (v) $x = 2.0\overline{5}$: $10x = 20.\overline{5}$, $100x = 205.\overline{5}$, $90x = 185 \Rightarrow x = \frac{37}{18}$.	1+1+1+1+1
	OR: (i) $\frac{1}{4} - \frac{5}{12} = \frac{3-5}{12} = -\frac{1}{6}$; $-\frac{1}{6} \div \frac{2}{3} = -\frac{1}{6} \times \frac{3}{2} = -\frac{1}{4}$. (ii) Sum $= \frac{6}{10} + \frac{7}{10} = \frac{13}{10}$; product $= \frac{21}{50}$.	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.